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HARVARD UNIVERSITY

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Academic Appointments

Harvard University (2023 – Present)
Postdoctoral Fellow, Environment and Natural Resources Program, Harvard Kennedy School Belfer Center & Salata Institute for Climate and Sustainability

Carnegie Mellon University (2022 – 2023)
Postdoctoral Associate, Department of Engineering and Public Policy
Visiting Researcher, Institute of Energy, Environment, and Economy, Tsinghua University (2023)

Education

Massachusetts Institute of Technology, Institute for Data, Systems, and Society
Ph.D. in Social and Engineering Systems, 2023 (defended September 2022)

- Dissertation: *Local official and polluter accountability in China's environmental inspections*

M.Fin., MIT Sloan School of Management (2017)

Zhejiang University
B.S. in Applied Mathematics (2011–2015)

Fields

Energy and Environmental Policy; Regulation and Policy Implementation
Methods: equilibrium modeling; causal inference; text-as-data methods

Publications

Policy Interactions and the Cost of Decarbonizing Aviation (first author, with Kristen McCormack, William A. Scott, Aaron Smith, Jingran Zhang, and James H. Stock) [link] [code]
Journal of the Association of Environmental and Resource Economists, forthcoming. NBER Working Paper No. 34326.
Policy impact: analysis of California's proposed SAF tax credit with this model was cited by the California Legislative Analyst's Office; covered by E&E News.

Dynamic Responses of SO_2 Pollution to China's Environmental Inspections (co-first author with Valerie J. Karplus) [link] [code]
Proceedings of the National Academy of Sciences, 2023

Working Papers

Credit Banks, Expectations, and the Cost of Layered Climate Policy: Evidence from U.S. Transportation Fuel
Three layers of climate policy apply to the same gallon of U.S. transportation fuel: a federal blending mandate with tradable credits, a federal production tax credit, and clean fuel standards in four states with banked credits and price caps. I ask what this stack costs and what one more federal dollar buys on top of it. I build a dynamic model of U.S. fuel and vehicle markets for 2026–2045 that solves the federal and state credit markets together, with investment estimated from 46 capacity decisions and credit-price expectations estimated from state credit markets. Current law abates carbon at about $134 a tonne and captures about 15% of the welfare gain of a carbon price at the social cost of carbon. Extending the production credit beyond 2029 is mostly a transfer. Because the federal mandate already sets the share of biofuel in fuel, the credit lowers the price of mandate credits rather than raising that share: about three quarters of each federal dollar reaches fuel buyers as cheaper fuel, capacity barely changes, and emissions rise by 64 million tonnes as cheaper fuel raises fuel use. Whatever EPA does with the mandate in response, the extension lowers welfare. State credit markets price far less scarcity than their statutes imply, so expectations change when these effects arrive but not their size. *Draft available on request.*

Research in Progress

Do State Low Carbon Fuel Standards Complement Federal Biofuel Mandates? (first author, with Kristen McCormack, William A. Scott, Aaron Smith, Jingran Zhang, and James H. Stock)
We study what state clean fuel standards add to federal biofuel mandates and how a state adopting one now should design it, in a model of fuel and vehicle markets with electric vehicle adoption and charging. In 2030, California's standard cuts fuel emissions in the state by about 32 million tonnes, but about three quarters of that cut is offset in other states as existing biofuel moves into California. It raises California's electric vehicle sales share from 21 to 33 percent. A new state that copies California's standard sits at its price cap unless it adds other credit types or a looser target. *To be presented at the APPAM Fall Research Conference, 2026.*

When Inspectors Return: Who Receives Substantive Enforcement?
I study whom local governments penalize when China's central environmental inspectors return. Using case records for more than 80,000 citizen complaints across two inspection rounds, I find that shutdown orders for verified complaints against small service businesses fell from 45 to 22 percent, compared with 52 to 39 percent for heavy industry. The relative decline is about 36 percent and appears in all ten provinces, while the share of complaints found valid did not change. *Presented at the CU Environmental & Resource Economics Workshop, 2026.*

Policy Writing

High Costs and Few Benefits from California's Proposed Sustainable Aviation Fuel Tax Credit (with Aaron Smith, Kristen McCormack, William A. Scott, James H. Stock, and Jingran Zhang) [link]
Energy Institute at Haas Blog, February 2026; cited by the California Legislative Analyst's Office.

Towards Credible and Cost-Effective Options for Sourcing Green Power for Supply Chains in China (with Da Zhang and Valerie J. Karplus) [link]
Policy brief for the Tsinghua Round Table on Corporate Carbon Neutrality Strategies, 2023.

Fellowships

Salata Institute for Climate and Sustainability Postdoctoral Fellowship, Harvard University, 2023–2027
MIT Presidential Fellowship, 2017

Teaching & Advising

Science, Technology and Public Policy, MIT (Spring 2020, Spring 2021)
Teaching Assistant for Professors Noelle Selin and Kenneth Oye; led weekly recitation sections (10–15 students), grading, and research-project mentoring.

The Economic Approach to Cities and Environmental Sustainability, MIT (Fall 2021)
Guest lecture: "Political Economy of Regulating Pollution"

MIT Kaufman Teaching Certificate Program, MIT (Fall 2020)
Designed course curriculum for "Data Science for Decision Making in Public Policy".

Research supervision, Harvard University (2026 – Present)
Supervise undergraduate research assistants on the clean-fuel policy modeling project.

Research supervision, Carnegie Mellon University (2022 – 2023)
Supervised master's research assistants on the Tsinghua green-power supply chain project (interview design, data collection, analysis).

Academic Talks

2027 ASSA Annual Meeting (poster, scheduled)

	2026	APPAM Fall Research Conference (scheduled); ENRP/STPP Fellows Seminar, Belfer Center (scheduled); Salata Institute Transportation Lab (scheduled); Tsinghua University; CU Environmental & Resource Economics Workshop; University of Hong Kong; Hong Kong University of Science and Technology; ARCS Research Conference; Renmin University of China; China and Energy Fellows Seminar, Belfer Center; Fudan University; Huazhong University of Science and Technology; Institutional and Organizational Economics Academy; CES NA Annual Conference
	2024	ASSA Annual Meeting
	2023	ARCS Annual Research Conference
	2022	Northeast Workshop on Energy Policy and Environmental Economics
	2020–21	Environmental Politics and Governance Conference; Duck Family Graduate Workshop in Environmental Politics and Governance (2020, 2021); Institutional and Organizational Economics Academy; ETH Academy on Sustainability and Technology; China Research Group Doctoral Symposium; Interdisciplinary Ph.D. Workshop in Sustainable Development
Policy & Industry Talks	2026	North American SAF Conference and Expo; International Fuel Ethanol Workshop & Expo
	2025	Regional SAF Workshop for Massachusetts and New England
	2024	North American SAF Conference and Expo
	2023	Roundtable on Corporate Carbon Neutrality Strategies, Tsinghua University

Service

Student organizer, Sloan China Seminar Series, MIT (2017–2020)
Research Collaborator, Sustainable PPE Initiative, MIT (2020–2021)

Languages

English (fluent), Chinese (native)

Software skills

Python, Stata, R, MATLAB, LaTeX

References

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